The University of New South Wales

School of Electrical Engineering and Telecommunications

ELEC3104: Digital Signal Processing

Final Examination, Session 1, 2017

Candidates should attempt ALL questions

1. Time allowed: 2 hours.

2. The paper contains **5** questions.

3. The questions are worth **105** marks in total, your final mark will be capped at **100**.

4. The questions are NOT of equal value. Part marks are as indicated .

5. The examination paper may **NOT** be retained by the candidate.

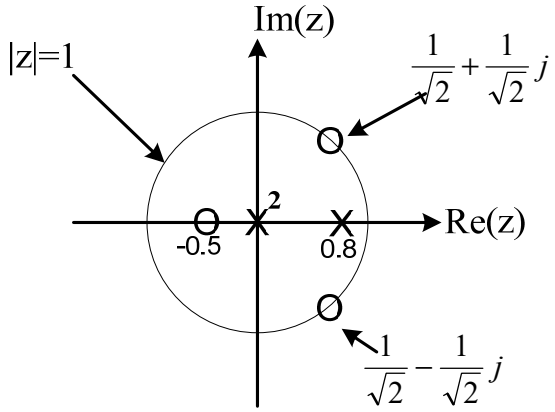
6. Authorised examination materials: ONE A4 paper with handwritten notes on both sides only.

7. This question booklet contains 3 pages apart from this instructions page

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

Question 1. (15 marks)

The pole-zero diagram of a system is given below.
The DC gain of the system is $15\left(1-\frac{1}{\sqrt{2}}\right)$.



(i) Sketch the approximate magnitude response of the system

(ii) Determine the transfer function, $H(z)$, of the system

(iii) Sketch the Direct Form I and Direct Form II implementations of this system

Question 2. (25 marks)

Design a discrete time low-pass IIR filter operating at a sampling rate of 8 kHz such that its magnitude response is monotonic (i.e., smooth with no ripples) and satisfies the following conditions:

(i) The magnitude response has an attenuation of at least 20dB at 2000 Hz

(ii) The magnitude response has an attenuation of at most 2dB at 1000 Hz

A. Determine the transfer function, $H(z)$.

[20 Marks]

B. sketch the Direct Form II structure of the filter.

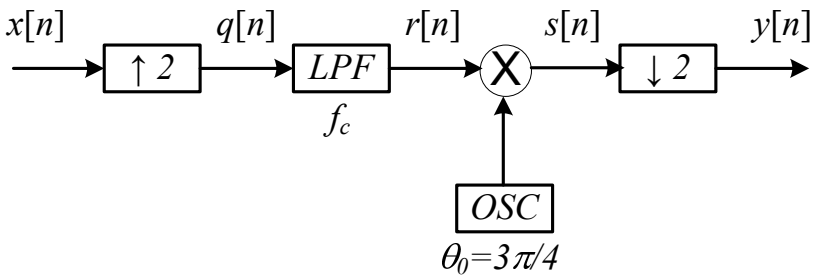
[5 Marks]

Use the Bilinear transformation to convert any analogue system to a discrete time system.

Question 3. (20 marks)

A discrete time high-pass filter operating at a sampling rate of 10 kHz and with a cut-off frequency of 2 kHz is required. The filter must also have linear phase characteristics. Design an 8th order FIR filter using the windowing method (using a rectangular window). Obtain the transfer function, indicating all the design steps, and sketch the Direct Form II implementation of the filter.

Question 4. (30 marks)



A. The oscillator (OSC) in the system shown above must have an impulse response given by, $h[n] = \sin(\theta_0 n)\, u[n]$.

(i) Determine the difference equation describing such an oscillator

(ii) Sketch its pole-zero plot

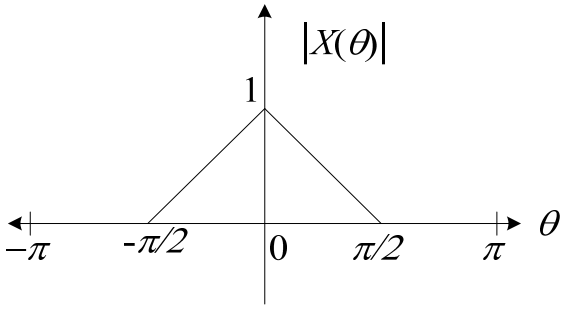
(iii) Draw its Direct Form II structure

[14 marks]

B. If the sampling rate of $x[n]$ is 500 Hz, what should be the cut-off frequency of the LPF (in Hz) such that $r[n]$ is an undistorted version of $x[n]$?

[2 marks]

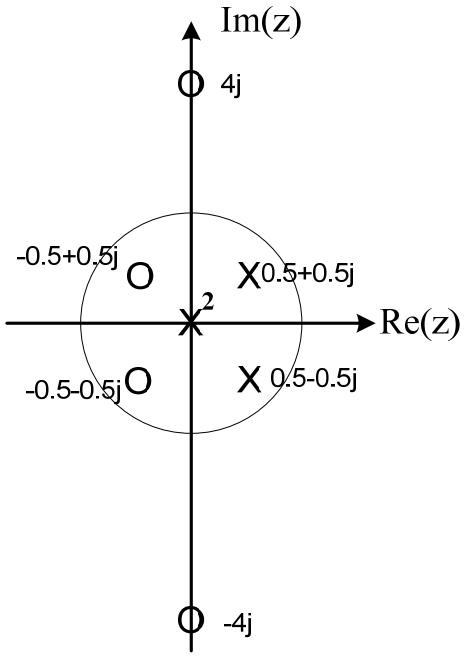
C. If $|X(\theta)|$, the magnitude spectrum of $x[n]$ is given as follows, sketch the magnitude spectra of $q[n]$, $r[n]$, $s[n]$ and $y[n]$. Use the same value as the cut-off frequency for LPF that you assumed in the previous question.



[14 marks]

Question 5. (15 marks)

The pole-zero diagram of a discrete-time system is shown below.



Implement this system as a cascade of a stable minimum phase filter and a stable all-pass filter.

A. Sketch the pole-zero plots of the minimum phase and all-pass filters

[6 marks]

B. Determine the transfer functions of the minimum phase and all-pass filters.

[5 marks]

C. Sketch the Direct Form II structures of both sub-systems.

[4 marks]

End of paper